MILITARY SPECIFICATION

MOLDING, PLASTIC, SHORT FIBER REINFORCED, STRUCTURAL

1. SCOPE

1.1 Scope. This specification covers the requirements for a short fiber reinforced structural molding plastic suitable for missile cases and other structural applications.

1.2 Classification. Materials provided to this specification shall be of one class.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-576	Boxes, Wood, Cleated, Veneer, Paper Overlaid
PPP-B-585	Boxes, Wood, Wirebound
PPP-B-591	Boxes, Fiberboard, Wood-Cleated
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock Corner
PPP-B-636	Boxes, Fiberboard
PPP-D-723	Drums, Fiber
PPP-D-729	Drums, Metal, 55 Gallon (For Shipment of Non-Corrosive Materials)

Military

MIL-B-117	Bags, Sleeves and Tubing, Interior Packing

STANDARDS

Federal

FED Test Method STD No. 406	Plastics, Methods of Testing

FSC 9330

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Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

HANDBOOKS

MIL-HDBK-5	Metallic Materials and Elements for Aerospace Vehicle Structures

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

D-256	Impact Resistance of Plastics and Electrical Insulating Materials, Test for
D-635	Flammability of Self Supporting Plastics, Test for
D-638	Tensile Properties of Plastics, Test for
D-648	Deflection Temperature of Plastics Under Load, Test for
D-695	Compressive Properties of Rigid Plastics, Test for
D-790	Flexural Properties of Plastics, Test for
D-792	Specific Gravity and Density of Plastics by Displacement, Test for
D-953	Bearing Strength of Plastics, Test for
D-1735	Water Fog Testing of Organic Coatings
D-2344	Apparent Horizontal Shear Strength of Reinforced Plastics by Short Beam Method, Test for
D-2648	Measuring Time-to-Failure by Rupture of Plastics Under Tension in Various Environments, Recommended Practice for

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19013.)

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(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies of the above publication should be addressed to the Uniform Classification Committee, 202 Chicago Union Station, Chicago Illinois. 60606)

INTERSTATE COMMERCE COMMISSION

49 CFR 71-90 Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles

(The Interstate Commerce Commission Regulations are now a part of the Code of Federal Regulations (latest cumulative packet supplement) available from the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402. Orders for the above publications should cite "Code of Federal Regulations, 49 CFR 71-90".)

MANUFACTURING CHEMISTS ASSOCIATION

Manual L-1 Guide to Precautionary Labeling of Hazardous Chemicals (Sixth Edition 1961)

(Application for copies should be addressed to the Manufacturing Chemists Association, Universal Building, 1825 Connecticut Avenue, Washington, D. C. 20009.)

3. REQUIREMENTS

3.1 *Material*. Material provided to this specification shall consist of chopped fiber strands which are impregnated with a suitable thermosetting resin.

3.1.1 Fiber. The reinforcing fiber shall be a chopped glass fiber strand of a length as specified by the procuring agency ±0.1 inch. It shall have a finish compatible with the resin selected, as demonstrated by an interfiber shear strength value of 8,000 psi or higher (see 3.5).

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3.1.2 Binder resin. The binder resin may be any thermosetting resin capable of meeting the requirements of this specification.

3.1.3 Form. The material may be a powder, pellets, plastic mass, uncured sheet stock, preform or otherwise as specified by the procuring agency (see 6.2).

3.1.4 Color. Unless otherwise specified by the procuring agency the color shall be natural (see 6.2). Natural colored materials are those to which no coloring materials of any sort have been added.

3.2 Physical and mechanical properties. Specimens prepared from the material in accordance with the manufacturer's instructions (3.6) shall conform to the requirements specified in tables I and II.

3.3 Creep. When tested in air with a relative humidity between 20 and 80 percent under a load of 50 percent of the mean tensile strength at 75 $\pm$5°F for 100 hours, the material shall not have a permanent elongation larger than 5 percent of the instantaneous deformation at that load. The test setup shall be, as described in ASTM D-2648.

3.4 Water absorption. The material when tested as specified in 4.6.1 shall not change weight more than 0.5 percent or change length more than 0.25 percent.

3.5 Interfiber shear strength. When tested as specified in 4.6.2, the shear strength between layers of fibers shall not be less than 8,000 psi. After 200 hours exposure to condensing humidity at 120 $\pm$5°F, the interfiber shear strength shall not be below 6,000 psi.

3.6 Manufacturer's instructions. The manufacturer shall provide complete instructions for using the material. Such instructions shall be furnished with each shipment of material and shall contain the following information:

a. Storage stability. The supplier shall provide the optimum temperature and packaging for reasonably maximizing the storage life at these conditions.

b. Molding techniques. The manufacturer shall provide the optimum "time-temperature-pressure-shape" and post cure cycles required to produce molded specimens conforming to the requirements of this specification.

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Table I. Mechanical properties of the plastic molding compound.

Property	Requirements At Temperature			ASTM Test Reference
	-65°F	75°F	270°F	
Tensile Strength ksi 1/	23	22	16	D-638
Tensile Modulus msi	4.0	3.5	3.0	D-638
Compressive Strength ksi 1/	35	35	20	D-695
Compressive Modulus msi	4.0	3.5	3.0	D-695
Impact Strength ft.lb./inch	30	28	25	D-256 Method A
Bearing Strength ksi (1.5 edge distance ratio)				D-953
4% Deformation	31	27	15	
Ultimate	35	25	17	
Interfiber Shear Strength ksi				See 3.5
Dry	--	8	--	
Wet	--	6	--	
Deflection Temperature Under Load (264 psi) °F	--	-	270	D-648
Flexural Strength ksi ksi 1/		50	---	D-790

1/ Strengths shall be "B" design allowables as determined by MIL-HDBK-5. (See 4.8)

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Table II. Physical properties of the plastic molding compound.

Property	Requirement	Test Reference
Specific Gravity	2.0 maximum	ASTM D-792
Resin Content	As specified by the procuring agency ±3% by weight	Federal Test Methods Standard No.406 Method 7061.
Volatiles Content	3% by weight maximum.	See 4.6.3

c. Mold shrinkage. The manufacturer shall provide the mold and post cure shirnkage.

3.7 Workmanship. The molding compound shall be uniform and free of contaminants, impurities and extraneous materials. No scrap material previously cured and reground shall be used.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that the material conforms to prescribed requirements.

4.2 Classification of examinations and tests. The examination and testing of the molding material shall be classified as follows:

(a) Material inspection (See 4.3)

(b) Quality conformance inspection (See 4.4).

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4.3 Material inspection.

4.3.1 Certificates of compliance. Material inspection shall be verified by certification for conformance to the requirements of this specification. When certificates of compliance are submitted, the Government reserves the right to check test the material to determine the validity of the certification. (See 6.3).

4.3.2 Inspection records. The suppliers inspection records of examination and tests for conformance to the requirements of this specification shall be kept complete and available to the purchaser upon request for a period of two years.

4.3.3 Lot. A lot of the material molding premix shall consist of all the material which is produced in one continuous manufacturing process and offered for acceptance at one time.

4.3.3.1 Inspection of acceptance of each lot shall consist of determining conformance to the requirements of this specification.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of sampling (4.4.1) and examination (4.5).

4.4.1 Sampling.

4.4.1.1 Sampling for visual examination. A sample of filled containers shall be selected from each lot in accordance with MIL-STD-105 at inspection level 1.

4.4.1.2 Sampling for tests. A sample of sufficient size to prepare the necessary specimens for testing shall be selected at random from each lot.

4.5 Examination.

4.5.1 Visual. Samples shall be visually examined for defects and at the acceptable quality level shown in table III.

4.5.2 Examination for delivery. Examination of material for packing and marking at the acceptable quality level shown in table III.

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Table III. Visual examination.

Item	A.Q.L Percent 1/	Classification of Defect	Defect 2/
Premix (see 3.1)	4.0	Major	Fiber length not as specified
Premix (see 3.1)	4.0	Minor	Color not as specified
Premix (see 3.7)	4.0	Minor	Reground scrap
Premix (see 3.7)	4.0	Minor	Contaminants foreign or extraneous material
In Open Container	4.0	Minor	Lack of, or improper liner in container
In Closed Container	1.0	Minor	Improper container
In Closed Container	1.0	Major	Improper weight (Any accurate scale) 3/
In Closed Container	1.0	Minor	Improper closure
In Closed Container	1.0	Minor	Lack of or improper marking

1/ Per MIL-STD-105.

2/ When applicable.

3/ The actual weight of a container filled with the minimum required quantity of premix shall be the basis for determining the acceptable weight of subsequent containers.

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4.6 Test procedures.

4.6.1 Water absorption. A cured sheet of the material shall be prepared approximately 0.1 inch thick. Ten specimens shall be cut approximately 1 inch by 2 inches in area. These specimens shall be measured and weighed, then immersed in clean water at 70 ±5°F for 200 ±5 hours. At the end of the soak period the specimens shall be removed from the water, blotted dry, let air dry not more than 15 minutes in air at 70 ±5°F and weighed. The average change in weight and length shall be determined. No specimen shall absorb more than 1% by weight water or change dimension more than 0.4 percent.

4.6.2 Interfiber shear strength. A unidirectional laminate panel shall be prepared using fiber sufficiently long to reach completely across the laminate panel. The fibers shall be the same fiber and shall have the same finish as the chopped fiber used in the molding compound. The resin shall be the same as the resin used in the molding compound and the time-temperature cycle shall be the same as that used for curing the molding compound. The panel when cured shall be approximately 0.1 inch thick.

4.6.2.1 Twenty horizontal shear specimens similar to those specified by ASTM D-2344 shall be cut out of the laminate with their long axis parallel to the fiber direction of the cured laminate. Ten of these specimens shall be exposed for 200 ±5 hours at 120 ±5°F to condensing humidity as defined by ASTM D-1735.

4.6.2.2 All the specimens shall be tested for interlaminar (interfiber) shear strength similarly to the test requirement of ASTM D-2344.

4.6.3 Volatiles content. Approximately 1 gram of the material shall be placed on a tared tray in a relatively thin layer and weighed to the nearest milligram (W_1).

4.6.3.1 The specimen on the tray shall then be placed in a ventilated forced circulation oven preheated to 200 ±10°F and held at that temperature for 60 ±10 minutes.

4.6.3.2 The specimen and tray shall then be removed from the oven, placed in a dessicator, let cool and then immediately weighed to the nearest milligram (W_2).

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4.6.3.3 The volatile loss shall be calculated in percent by weight volatiles lost.

$$\text{Percent by weight volatiles lost} = \frac{W_1 - W_2}{W_1} \cdot (100)$$

where

W_1 = the original specimen weight

W_2 = the weight of the specimen after drying and cooling.

4.7 Rejection and test

4.7.1 Visual inspection. A lot shall be rejected when the number of nonconforming units of a sample exceeds the acceptance number for that sample when examined in accordance with 4.5 and table III.

4.7.2 Test. A lot shall be rejected for failure to comply with any of the test requirements of this specification when tested in accordance with tables I and II.

4.8 Reduction of strength data to "B" design allowables. At least ten specimens shall be measured for each of the tensile, compression and flexural strength requirements of table I. The standard deviation of the data shall be calculated as follows:

$$S^2 = \sum_{i=1}^{N} \frac{(X_i - \overline{X})^2}{N-1}$$

where:

N is the number of specimens measured

X_i is a test value

$$\overline{X} = \frac{1}{N} \sum_{i=1}^{N} X_i$$

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The "B" design allowable shall then be determined from

$$F_B = \overline{X} - K_B S$$

where:

K_B is a reduction factor determined from statistical factors which are provided graphically by MIL-HDBK-5 or the USAF DESIGN GUIDE FOR ADVANCED COMPOSITES

N	K_B
5	3.50
10	2.35
15	2.20
20	1.18

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. The preservation, packaging, and packing requirements specified herein apply only to direct purchases by or direct shipment to the Government.

5.1.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1.1 Powder, pellets or bulk premixed molding. (Not applicable).

5.1.1.2 Uncured sheet stock

5.1.1.2.1 Level A. The sheets shall be wrapped individually or interleaved with polyethylene film and packaged in bags conforming to MIL-B-117, Type I, Class C. Where flexibility permits wrapping on 4 inch diameter cylindrical cores and shipment in rolls is permitted. Rolled sheets shall be packaged in bags conforming to MIL-B-117, Type I, Class C and individually packaged in weather resistant class fiberboard boxes in conformance with PPP-B-636, in such fashion that the roll is suspended by the ends of the core.

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5.1.1.2.2 Level C. Preservation and packaging shall be sufficient to afford adequate protection against damage and deterioration during shipment. This may conform to the industries best commercial practice when such meets the requirements of this level.

5.1.1.3 Preforms. Preforms shall be supported as specified by the procuring agency and packaged similarly to uncured sheet stock as specified in level A or C.

5.2 Packing

5.2.1 Packing. Packing for powder, pellets or bulk premixed molding compound and uncured sheet stock shall be level A or C as specified.

5.2.1.1 Powder, pellets or bulk premixed molding

5.2.1.1.1 Level A. This material shall be packed in one of the containers listed below. The gross weight shall not exceed 200 pounds. All drums shall be furnished with a polyethylene bag properly closed. Closure of drums shall be in accordance with the applicable specification.

Fiber drums	PPP-D-723, Type III, Grade A
Fully removable head metal drums	

5.2.1.1.2 Level C. The material shall be packed for shipment in compliance with the industries best commercial practice where such meets the requirements of this level.

5.2.1.2 Uncured sheet stock

5.2.1.2.1 Level A. Stocks of sheets packaged as specified in 5.1 shall be packed in boxes conforming to any of the following specifications at the option of the supplier.

PPP-B-576	Class 2
PPP-B-585	Class 3 Use
PPP-B-591	Overseas Type
PPP-B-601	Overseas Type
PPP-B-621	Class 2
PPP-B-636	Class 2

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Rolled materials packaged as specified in 5.1 shall be packed in export type shipping containers conforming to

PPP-B-576
PPP-B-585
PPP-B-591
PPP-B-601
PPP-B-621
PPP-B-636

5.2.1.2.2 Level C. The materials shall be packed for shipment in compliance with the industries best commercial practice.

5.2.1.3 Preforms. Preforms packaged as specified by 5.1 shall be packed similarly to uncared sheet stock materials.

5.2.2 Exterior containers. As far as possible, exterior containers shall be of uniform shape and size shall be of minimum cube and tare consistent with the protection required, and shall contain identical quantities. The gross weight of each pack shall be limited to approximately 200 pounds.

5.2.2.1 Container closing. Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto.

5.3 Marking for shipment. In addition to any special marking required by the contract or purchase order (6.2) or herein, interior packages, exterior shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-129. The nomenclature shall include:

a. MOLDED, PLASTIC, SHORT FIBER REINFORCED, STRUCTURAL

b. Specification number

c. Form (powder, pellets, compound, sheet or preform)

d. Length of reinforcement

e. Color (if specified)

f. Resin content (wt. %)

g. Manufacturer's name and Code No.

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h. Date of manufacture

i. Quantity

5.3.1 Hazardous chemicals. All packaged units of hazardous chemicals shall have affixed thereto such warning labels and markings as may be required by the Interstate Commerce Commission CFR, Title 49, Part 71-90, and the Manufacturing Chemists Association Manual L-1.

5.3.2 Warning markings. When applicable, packaged units shall be marked essentially as follows:

"WARNING: USE WITH ADEQUATE VENTILATION - FLAMMABLE

The solvents contained in this compound may have flash points well below prevailing room temperatures. Do not heat by, or use near, an open flame. Heat preferably by means of a steam coil or other nonflame-producing heat sources."

"WARNING: Curing agents, accelerators, and solvents to be used with this compound should be stored in original tight containers isolated from one another in cool locations. Never permit a catalyst to come in contact with any component except the base compound."

5.4 Instruction sheet. Unless otherwise specified, a copy of the latest approved issue of the supplier's instruction sheet shall be included with each interior package (see 3.6).

5.5 Air transport. When specified in the procurement documents (6.2), preparation for delivery shall be modified to meet any special requirements of air transport.

6.1 Ordering data. The procuring documents should specify the following:

a. Title, number and date of this specification.
b. Fiber length required.
c. Form.
d. Color, if required.
e. Instruction sheet.

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f. Level of packaging and packing required.
g. Special marking, if required.
h. Size and type of containers required.
i. Whether palletizing is required.
j. Special preparation for delivery requirements for air transport, if applicable.

6.3 Certification. Government as referenced in 4.3.1 is the Air Force Materials Laboratory, ATTN: MXE, Wright-Patterson Air Force Base, Ohio, 45433.

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